Literature Review for Master Thesis

1 Avalanche Modelling

1.1 Numerical investigation of crack propagation regimes in snow fracture experiments (Bobillier et al., 2024)

1.2 Micromechanical modeling of snow snow failure (Bobillier et al., 2020)

1.3 Modeling snow slab avalanches caused by weak-layer failure – Part 2: Coupled mixed-mode criterion for skier-triggered anticracks (Rosendahl & Weißgraeber, 2020)

1.4 Investigating the release and flow of snow avalanches at the slope-scale using a unified model based on the material point method (Gaume et al., 2019)

1.5 Numerical investigation of the mixed-mode failure of snow (Mulak & Gaume, 2019)

1.6 Snow fracture in relation to slab avalanche release: critical state for the onset of crack propagation (Gaume et al., 2017)

1.7 Modeling of crack propagation in weak snowpack layers using the discrete element method (Gaume et al., 2015)

Main Question of article: Which snow parameters influence crack propagation distance and speed and how can these processes be modeled?
Why this is relevant for the thesis: Understanding the influence of different snow parameters on crack propagation distance and speed is important to model and understand slab breakoff DAS data in thesis.

Gaume et al., 2015 state that dry-snow slab avalanches form by the failure of a weak snow layer underlying a cohesive slab. The local damage leading to failure is a crack in the weak layer, which when it reaches critical length leads to tensile fracture in the slab, which leads to avalanche release (Gaume et al., 2015). Understanding how these processes can be modeled and which parameters affect the propagation distance and speed is important for the thesis because slab breakoff is investigated in DAS data. Understanding which snow parameters lead to large crack propagation or speed is important to determine in which DAS datasets slab breakoff is most likely to be detected and at which times in the dataset this would be (does the breakoff take a lot of time or not because speeds and distances are low or high?). Gaume et al., 2015 state that the dynamic phase of crack propagation and the effects of different snow parameters are poorly understood, which limits avalanche forecasting because it is unclear how and under which circumstances cracks propagate. This is directly linked to the thesis, because cracking of the snow layer is investigated in DAS data. The method used by Gaume et al., 2015 to model cracking in snow is the discrete element method described by Cundall and Strack, 1979, with a limitation that the slab is modeled as a continuos material with elastic-brittle behavior.

The loading modeled is due to gravity and the propagation saw test (Gaume et al., 2015), the cohesive part of the snow (so the connections between the grains) are modeled by the linear contact law, the maximum and minimum and minimum tensile and shear stresses are modeled by beam theory (Gaume et al., 2015). The stiffness at the bonded contact (of grains) is calculated (Gaume et al., 2015), this is important to see how the material reacts: what influence does the stiffness of the grain contacts have on crack propagation distance and speed? Gaume et al., 2015 calculate the time-step of the simulations from the snow grain properties, this ensures that the maximum crack propagation velocity is lower than the speed of elastic waves in the snow, this means that the cracks in the snow cannot propagate faster than the propagation of the elastic waves which are caused by the loading. Understanding the relationship between elastic wave velocity and crack propagation speed is also important for the modeling of DAS data of slab breakoff which will be done in the thesis.

Gaume et al., 2015 determined the propagation distance of cracks in the slab by setting the tensile and strength to finite values, to determine propagation speed on the other hand, these values were set to infinity. Understanding the propagation speed and distance of cracks is important because the higher either of these values, the faster the avalanche breaks off. The propagation distance described by Gaume et al., 2015 is defined as the distance between the point where failure initiates, so the point where the initial load is applied and the point where the tensile failure is located. The numerical models showed that the crack propagation speed increases with increasing young's modulus, increasing thickness of slab and increasing density, but the thickness of weak layer has no influence on propagation speed (Gaume et al., 2015). From their results, Gaume et al., 2015 suggest that the lower the young's modulus of a snow layer, meaning the softer a snow layer, the lower the crack propagation velocity, which means that avalanches break off less fast. This suggests that the propagation speed of cracks is mainly influenced by failure conditions such as the tensile strength of the snow layer rather than structural parameters such as weak layer thickness (Gaume et al., 2015), which is relevant for the thesis because factors such as weak layer thickness need not be modeled, which saves computational time. The propagation distance of cracks is influenced by weak layer thickness however, Gaume et al., 2015 suggest that if weak layer thickness increases, there is more slab bending which leads to a higher tensile stress and more likely tensile failure, which means that crack propagation distance decreases, because failure happens before the cracks can propagate very far (Gaume et al., 2015). Furthermore, the simulations run by Gaume et al., 2015 show that the slope angle has little influence on crack propagation for low tensile strength snow layers, but for high tensile strength layers there is a large correlation between slope angle and propagation distance of cracks (Gaume et al., 2015). This can be explained by the main failure mode being tensile failure due to slab bending at high tensile stresses of the snow layer (Gaume et al., 2015).

Gaume et al., 2015 conducted lab experiments, which confirm the results of the numerical simulations in all but one case: the relationship between high young's modulus and large crack propagation is counter-intuitive, because in harder snow, cracks propagate less well compared to soft snow, Gaume et al., 2015 explain this by the limitations of the beam theory to account for dynamic effects. This is an important consideration depending on which method will be used to model crack propagation in the DAS data. Gaume et al., 2015 show that the stress evolution in the snowpack always shows an increase of tensile stress at the top of the slab before crack propagation, which happens close to the tip of the crack in the weak layer. The tensile crack opens up as soon as the tensile stress is equal to the tensile strength of the snow, this crack always starts at the top surface of the slab (Gaume et al., 2015). Understanding the stress evolution of the snowpack is important for DAS data interpretation, because it is possible that the different phases of this path (increase in stress, opening of crack in weak layer, tensile cracking and failure) can be seen in the data. Gaume et al., 2015 mention two important limitations of the simulation results: the models assume a simplified shape of the weak layer, as well as the assumption that the crack propagation speed is lower than the speed of the elastic wave. At thigh propagation speeds of the crack, the tangential displacement of the snowpack becomes more important than the vertical displacement (Gaume et al., 2015). Furthermore, the coulumn length of the modeled propagation saw test, which results in loading of the modeled snow, has large influences on the simulation results depending on the snow properties (Gaume et al., 2015), the thicker and stronger the snow layer, the larger the column must be to get accurate results.

To summarize the snow parameters important for propagation distance and speed, which are important for simulations of slab breakoff in the thesis:

- propagation speed increases if: young's modulus increases, slab depth increases, slab density increases or the slope angle increases
- propagation velocity decreases if weak layer has increasing strength
- propagation distance decreases if young's modulus, slab density or weak layer thickness increases (leads to faster failure)
- propahation distance increases if slab tensile strength, slab depth, weak layer strength or slope angle increases

1.8 A physical SNOWPACK model for the Swiss avalanche warning: Part I: numerical model (Bartelt & Lehning, 2002)

Main Question of article: How can the snowpack and its governing equations be modeled accurately but as simple as possible?
Why this is relevant for the thesis: Modeling snowpack evolution and structure is important to forecast avalanches (where are avalanches likely to form), but also to understand avalanche dynamics (in which layers do avalanches form)

Bartelt and Lehning, 2002 describe that numerical modeling of the snowpack is important for avalanche prediction. The snowpack-model developed solves the partially governing equations using finite-element methods (Bartelt & Lehning, 2002). Understanding snowpack evolution and avalanche formation is also relevant for the thesis because the crack propagation which is modelled in the thesis depends on snow properties. According to Bartelt and Lehning, 2002, most avalanches occur during or after periods of intense snowfall. Snow settling gives an insight on mechanical stability of snow, which is also important to understand crack propagation: in stable layers cracks probably do not propagate as well as in layers which already have some instabilities. The microstructure and the layering of snow also needs to be modeled to detect discontinuities or layer boundaries, which is important to assess stability of layers (Bartelt & Lehning, 2002). Bartelt and Lehning, 2002 also model the influence of solar radiation and warm winds on the energy exchange, the melting of snow and the subsequent transport of meltwater is important to understand the release of wet snow avalanches.

Existing snowpack models such as CROCUS rely on Trusdall's equation for mixture and postulate conservation of ice, water, water vapor and air of the snowpack, these models are simplified (Bartelt & Lehning, 2002). This means that the existing models for example do not describe settling of the snowpack, they assume the snowpack to be dry or the absence of snow metamorphism, which does not depcit the reality of the snowpack accurately (Bartelt & Lehning, 2002). An accurate and physical model of the snowpack is important for the thesis because modelled data should be comparable to field data measured by the DAS cable. This would not be the case if simplifications are made in the simulation of crack propagation in snow, the simulated data would look different from the measured data. The novel SNOWPACK model described by Bartelt and Lehning, 2002 uses three constituents (ice, water and moist air) to describe the snow. These three all have the same temperature, which is the snow bulk temperature measured in meterological observations, assume all meltwater processes to be energy and mass conserving and assume that the weight of the snowpack is carried by a single ice lattice (Bartelt & Lehning, 2002).

Bartelt and Lehning, 2002 use four governing equations to describe the snowpack: the energy transfer equation at the surface (conservation of mass of vapor and water and the bulk temperature), because all of the snow constituents have the same temperature, a momentum equation for settling of the snow because the weight of the snow is carried by a single ice lattice: The conservation of mass of ice is not considered because the numerical solution of the displacement is continuos because of the lagrangian coordinate system. Any change in the snow load results in a change of the stress state of the settlement equation. Snow cannot be assomed to be a plastic material because deformations caused by loading are irreversible, understanding this is important for crack formation and propagation. Bartelt and Lehning, 2002 decsribe a microstructure-based viscosity formulation, plasticity theory is not applicable because of time-dependent yield surfaces and a directionality of plastic flow which is defined in relation to the yield surface. The initial conditions of the snowpack such as snow height are taken from meterological data, the energy exchange at the surface is the most important boundary condition (Bartelt & Lehning, 2002). Neumann boundary conditions are used if the temperature at the surface is close to the melting temperature, if the temperature is well below the melting temperature, Dirichlet boundary conditions are used (Bartelt & Lehning, 2002).

The melting and refreezing processes in the snowpack are essential in modelling snow structure changes, which have an effect on DAS signal propagation asn well as avalanche formation. Bartelt and Lehning, 2002 model these by the numerical heat transfer solution. Because the snow temperature is assumed to be constant 0°C, if the heat transfer solution is greater than 0°C, the excess energy is assumed to be used for melting, if the heat transfer is smaller than 0°C, the energy is assumed to be used for refreezing (Bartelt & Lehning, 2002). Bartelt and Lehning, 2002 used the snow structure measurements of winter 1999, which was a winter with numerous avalanches, to verify the SNOWPACK model developed. Generally, there is good agreement between the measured and modelled data, which means that the model is physical, with some exceptions (Bartelt & Lehning, 2002). If the calculated snow height is different from the measured snow height, the model corrects this by adding elements to the finite element grid in case the calculated height is too low, or removing elements if snow hieght is overestimated (Bartelt & Lehning, 2002). This leads to mistakes if the snowpack settles quickly: if there is quick settlement, snow height decreases quickly because of densification, the model would remove elements at the surface without accounting for densification of the snow at the bottom (Bartelt & Lehning, 2002). Snow density and snow settlement have strong effects on heat conductivity of snow, Bartelt and Lehning, 2002 mention that errors in either of these fields have strong effect on the entire simulation.

The SNOWPACK model developed by Bartelt and Lehning, 2002 includes modeling of the microstructure of snow as well as density discontinuities, which represent layer boundaries, which is important to understand snow structure as well as crack propagation related to avalaches. The multicomponent description developed allows for representation of mass and energy conservation (Bartelt & Lehning, 2002), which can be used to asess snow metamorphism related to temperature and moisture, but also to determine where wet snow avalanches form due to the presence of water. Bartelt and Lehning, 2002 desrcibe the volumetric deformation of fresh snow by large strain description, the thermal boundary conditions which need to be used can be determined by meterological data. These data can also be used to determine if there is melting or refreezing of snow, which again is important for snow structure and stability. The laws for heat conductivity and viscosity related to microstructure show the relations between densification of snow and heat transfer (Bartelt & Lehning, 2002).

1.9 A numerical model to simulate snow-cover stratigraphy for operational avalanche forecasting (Brun et al., 1992)

Main Question of article: How do snow conditions such as temperature gradient and water content influence snow metamporphism and how can this be implemented in the CROCUS-snowpack model?
Why this is relevant for the thesis: Snow metamorphism changes grain properties, which is important for the formation of weak layers which can trigger avalanche release, but also overall stability of the snowpack. Furthermore, layers of different grain properties will depict DAS signals differently.

When a snowpack is subjected to a change, like a change in temperature or pressure conditions, the snow grains change (Brun et al., 1992). Snow metamorphism changes the grain sizes and shapes, which changes mechanical stability of the snow (Brun et al., 1992). This is relevant for avalanche modelling because certain types of snow metamorphism can weaken layers in snow, such layers can act as failure planes and make release of avalanches easier. Snow metamorphism is affected by the water content of the snow, the capillary pressure of the water between the snow grains, the radius of curvature of the grains (shape), as well as the temperature gradient (Brun et al., 1992).

Brun et al., 1992 describes the main limitation of past methamorphism models as assuming the snowpack to have homogenous shapes and sizes, which are often spherical or simplified geometric shapes, this does not represent snow accurately however, especially fresh snow is very heterogenous. Brun et al., 1992 conducted experiments for different snow types and conditions to determine new equations to describe metamorphism. These experiments showed that the rate and type of metamorphism does not depend on crystal type but rather on the temperature gradient (Brun et al., 1992). A gradient below 5°C/m results in rounded crystals, a gradient above 5°C/m results in facteted crystals, which transform to depth hoar, if a gradient of more than 15°C/m is applied (Brun et al., 1992). This is relevant for avalanche modelling, because it is critical to understand how snow grains evolve to asess their stability, which can be used to determine where weak layers will form. These experiments were conducted for dry and wet snow conditions.

Brun et al., 1992 describe a new description of snow, which takes into account the dendricity, sphericity and size of the grains. When snow falls, new layers with different properties are added to the snowpack (Brun et al., 1992). Simulating these different layers is again important to assess the stability as well as interactions of the different layers. Brun et al., 1992 combined the observations made in the laboratory with the CROCUS snowpack model to simulate these different layers based on meterological observations. Using MEPRA (a system for avalanche risk forecasting), the shear stress of the different layers was estimated at a 45° angle. This is important for avalanche forcasting, because the strength of different layers can be assessed and it can be determined under which stress the layers fail.

1.10 A discrete numerical model for granular assemblies (Cundall & Strack, 1979)

Main Question of article: How can the forces and the resulting displacements be modeled in a granular medium?
Why this is relevant for the thesis: Snow consists of grains, forces at the surface such as mechanical loading can cause displacement of the individual grains. If this happens at a very large scale, it is an avalanche.

The distinct element method described by Cundall and Strack, 1979 is a numerical modeling method to model the mechanical behavior of a material composed of disks spheres. This is relevant to the thesis because snow is composed of individual snow grains, which interact with each other at their contacts, all the grains together form the snow pack. It is important to model the behavior and displacement of the grains to model avalanche dynamics because avalanches form through loading or failure in a weak layer, which then propagates through the entire snowpack (Gaume et al., 2017).

Since a snowpack is a medium composed by individual snow grains, which are not entriely compacted (contain air between the grains) and interact with each other at grain contacts, the medium described by Cundall and Strack, 1979 represents snow well. It is a granular medium composed of distinct particles, which interact with each other at their contacts and form a porous medium (Cundall & Strack, 1979). The loading and unloading of such materials is complex and stresses cannot be measured easily because of the interactions of different grains (Cundall & Strack, 1979).

The distinct elememt method can model particles of any shape (Cundall & Strack, 1979), which is important for snow modelling because the snow grains are not always spherical. Cundall and Strack, 1979 model the transient interaction of particles by calculating the contact forces and displacements of the stressed discs (particles) caused by disturbances at the boundaries. In the avalanche case discussed in the thesis, the distrubance is represented by loading of the surface, which can be caused by a skier, which causes failure in a weak layer, or cracking in the snowpack. This results in stress propagation through the snowpack, which results in displacement of the particles, which is the avalanche motion. The resultant foces on the disks are calculated only by the discs which are in direct contact (all the disks that touch one disk cause its resultant force) (Cundall & Strack, 1979). Cundall and Strack, 1979 describe a model only for dry materials, this means this is only applicable for dry snow avalanches which contain no water between the snow grains.

Numerical modelling, such as described by Cundall and Strack, 1979is advantageus over lab tests and experiments because it is less time consuming and any data can be accessed at any stage, which means that parameters such as loading, particle size, and properties such as stiffness and shape can be changed at any stage. In the avalanche case, this means that simulations can be run for different snow properties and loading scenarios, for example a simulation for older, more compacted snow can be run as well as a simulation for fresh snow. The numerical model for spheres is based on methods described by Serrano and Roderiguez-Ortiz (1973, I CANNOT FIND THIS PAPER ONLINE). The forces at the interfaces between the spheres are calculated by incremental displacements of the centers of particles, where shape changes of the individual particles are neglible (Cundall & Strack, 1979). How can the center of the grain be determined if the grain has an irrgular shape? The matrix representing contact stiffness between the grains described by Cundall and Strack, 1979 needs to be recalculated whenever a new contact is made or broken, meaning when there is a crack in the snowpack or when new contacts between snow grains are made. The calculations of forces are based on Newtons 2nd law, which describes the movement of a particle as the result of the forces acting on it. The deformation is neglected because the deformation of individual particles is small compared to the deformation of the medium as a whole (Cundall & Strack, 1979). This means that the deformation of the individual snow grains is small compared to the deformation of the entire snowpack which represents the avalanche movement. What about the fusing / compaction of grains which changes snow properties? The relative displacement of a particle is calculated by the sum of the force increments acting on it, partciles can also overlap, in this case the contact foces and the interaction forces also need to be calculated (Cundall & Strack, 1979). The location of the rigid boundaries (walls) described by Cundall and Strack, 1979 needs to be specified before modelling. This could be a possible limitation for the avalanche case because there is only one rigid ”wall” which is the interface of snow and the rock below it.

2 Crack Propagation

3 Relevant Source Mechanisms

4 Bibliography

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